

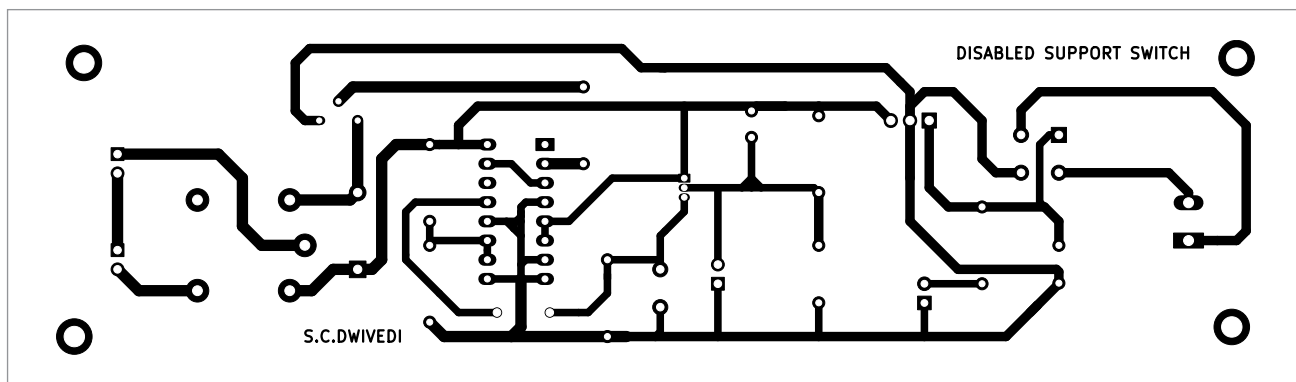


Fig. 3: Actual-size PCB layout for the circuit

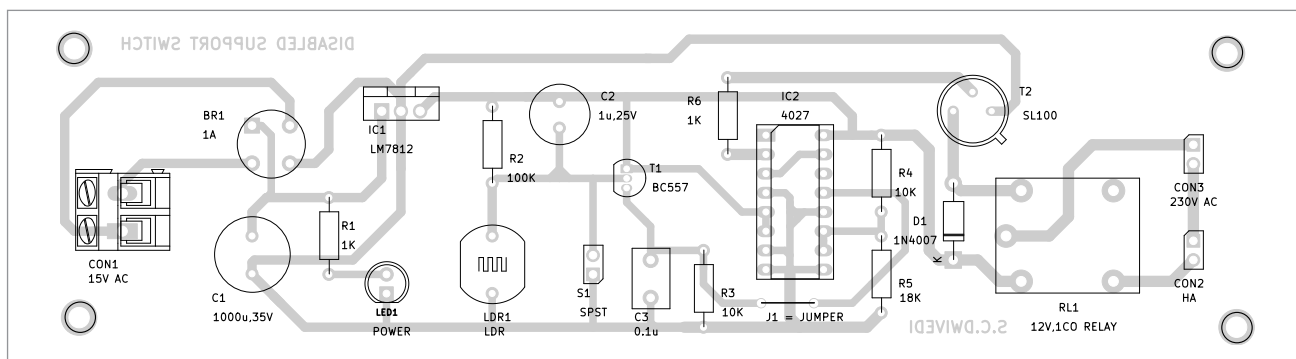


Fig. 4: Component layout of the PCB

PARTS LIST

Semiconductors:

IC1	- LM7812, 12V voltage regulator
IC2	- 4027 dual flip-flop IC
BR1	- 1A bridge rectifier
T1	- BC557 PNP transistor
T2	- SL100 NPN transistor
D1	- 1N4007 rectifier diode
LED1	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R6	- 1-kilo-ohm
R2	- 100-kilo-ohm
R3, R4	- 10-kilo-ohm
R5	- 18-kilo-ohm

Capacitors:

C1	- 1000 μ F, 35V electrolytic
C2	- 1 μ F, 25V electrolytic
C3	- 0.1 μ F ceramic disc

Miscellaneous:

LDR1	- Light dependent resistor (LDR)
RL1	- 12V, 1CO relay
S1	- Foot switch
X1	- 230V AC primary to 15V, 500mA secondary transformer
CON1-CON3	- 2-pin terminal
	- Torch or laser light

output goes high at its pin 2, which energises relay RL1 via relay driver transistor T1. As a result, the home appliance gets 230V AC supply and switches on. When the light is focused once again on the LDR, it re-triggers flip-flop IC2 and now its output goes low at its pin 2 and relay RL1 de-energises via transistor T1. As a result, the home appliance now gets disconnected from 230V AC and switches off.

Similarly, when a hands-disabled person presses foot switch S1, it triggers flip-flop IC2 and its output goes high at its pin 2, which energises relay RL1 via relay driver transistor T1. As a result, the home appliance gets 230V AC and switches on. When switch S1 is pressed again, it re-triggers flip-flop IC2 and its output goes low at its pin 2, which de-energises relay RL1 via transistor T1. As a result, the home appliance gets disconnected from 230V AC and switches off.

Construction and testing

An actual-size, single-side PCB for the switch is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on PCB, enclose it in a suitable box. Connect LDR in front of the box and extend the foot switch to the disabled person's chair using a flexible wire.

After proper installation, connect 230V AC mains to primary terminals of the transformer and 15V AC secondary terminals across CON1. The circuit is now ready to use, provided the LDR has been installed at the domestic switchboard and the foot switch has been extended up to the disabled person's chair. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/disabled-support-switch>

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